SRM Institute of Science and Technology

Tiruchirappalli



TECHNICAL REPORT

Computer Networks Lab Test

Course Code: 21CSC302J

Academic Year: 2026-2027

Faculty

Engineering and Technology

Department

CSE - Cyber Security

Programme

B. Tech

Semester

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Academic Year

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Course Code & Title

21CSC302J – Computer Networks

Faculty Handling

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EXPERIMENT 1: Basic network connectivity using Switch

1. Aim

To establish communication between two PCs connected through a network switch using Cisco Packet Tracer and verify connectivity using the ping command.

2. Software Requirements

- Cisco Packet Tracer
- Two PC-PT devices
- One Cisco 2960-24TT switch
- Copper straight-through Ethernet cables

3. Network Topology

Two PCs, PC0 and PC1, are connected to a Cisco 2960-24TT switch using Ethernet cables.

Topology:

PC0 ↔ Switch0 ↔ PC1

4. Procedure

1. Open Cisco Packet Tracer.
2. Place two PC-PT devices and one 2960-24TT switch in the logical workspace.
3. Connect PC0 and PC1 to the switch using copper straight-through cables.
4. Configure the IP addresses of both PCs in the same subnet.
    - **PC0 IP Address:** 192.168.1.1
    - **PC1 IP Address:** 192.168.1.2
    - **Subnet Mask:** 255.255.255.0
5. Open PC0 and navigate to **Desktop → Command Prompt**.
6. Execute the command:

ping 192.168.1.2

7. Observe the replies received from PC1 and check the ping statistics.

5. Command Used

C:\> ping 192.168.1.2

6. Observation and Result

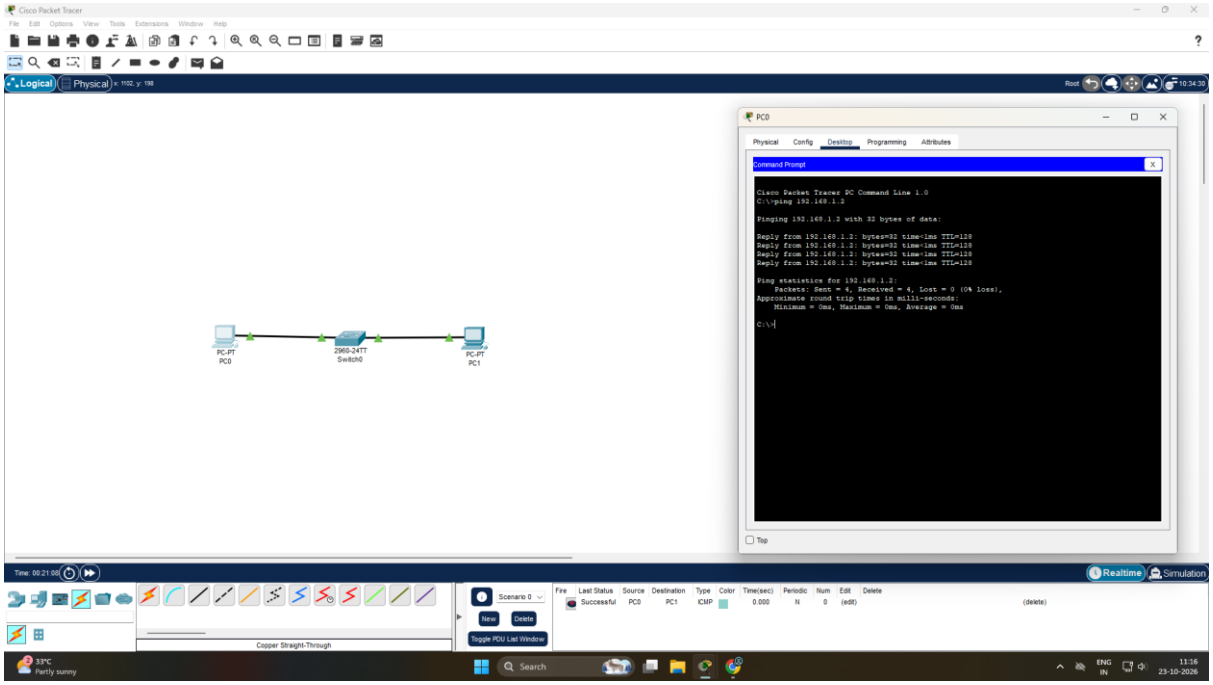
The ping command returned successful replies from IP address 192.168.1.2.

The observed statistics were:

- Packets sent: 4
- Packets received: 4
- Packets lost: 0
- Packet loss: 0%
- Approximate round-trip time: 0 ms

All four ICMP echo requests received replies, indicating successful communication between the two PCs.

Screenshot:



7. Conclusion

Communication between PC0 and PC1 was successfully established through the Cisco 2960-24TT switch. The successful ping test with 0% packet loss confirms that the devices can communicate over the configured local area network (LAN).

EXPERIMENT 2: Basic Router Configuration and password Security

1. Aim

To perform basic configuration of a Cisco router using Cisco IOS command-line interface (CLI) commands, configure passwords for privileged EXEC mode and console and virtual terminal access, enable password encryption, and save the configuration.

2. Software Requirements

- Cisco Packet Tracer
- One Cisco 2911 router
- Two PC-PT devices
- Copper straight-through Ethernet cables

3. Network Topology

Two PCs, PC0 and PC1, are connected to the interfaces of a Cisco 2911 router.

Topology:

PC0 ↔ Router0 ↔ PC1

4. Procedure

1. Open Cisco Packet Tracer.
2. Place two PC-PT devices and one Cisco 2911 router in the workspace.
3. Connect PC0 and PC1 to the router using copper straight-through Ethernet cables.
4. Open Router0 and select the **CLI** tab.
5. Enter privileged EXEC mode.
6. Enter global configuration mode and configure the router hostname.
7. Set an enable secret password.
8. Configure passwords for console access and virtual terminal lines.
9. Enable password encryption.
10. Exit configuration mode and save the running configuration to startup configuration.

5. Commands Used

The following Cisco IOS commands were entered in the router CLI:

Router> enable

Router# configure terminal

Router(config)# hostname Router0

Router0(config)# enable secret cisco

Router0(config)# line console 0

Router0(config-line)# password cisco

Router0(config-line)# login

Router0(config-line)# exit

Router0(config)# line vty 0 4

Router0(config-line)# password cisco

Router0(config-line)# login

Router0(config-line)# exit

Router0(config)# service password-encryption

Router0(config)# end

Router0# copy running-config startup-config

When prompted for the destination filename, press **Enter** to accept the default filename.

6. Explanation of Commands

Command	Purpose
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode.
hostname Router0	Sets the router's hostname to Router0.
enable secret cisco	Configures the encrypted privileged EXEC password.
line console 0	Enters console line configuration mode.
password cisco	Sets the password for the selected access line.
login	Requires password authentication on the selected line.
line vty 0 4	Configures virtual terminal lines 0 through 4.
service password-encryption	Encrypts supported plaintext line passwords in the configuration.
end	Returns to privileged EXEC mode.
copy running-config startup-config	Saves the current configuration for use after a restart.

7. Observation and Result

The router accepted the configuration commands. The hostname was set to Router0, the enable secret and line passwords were configured, password encryption was enabled, and the configuration was saved successfully.

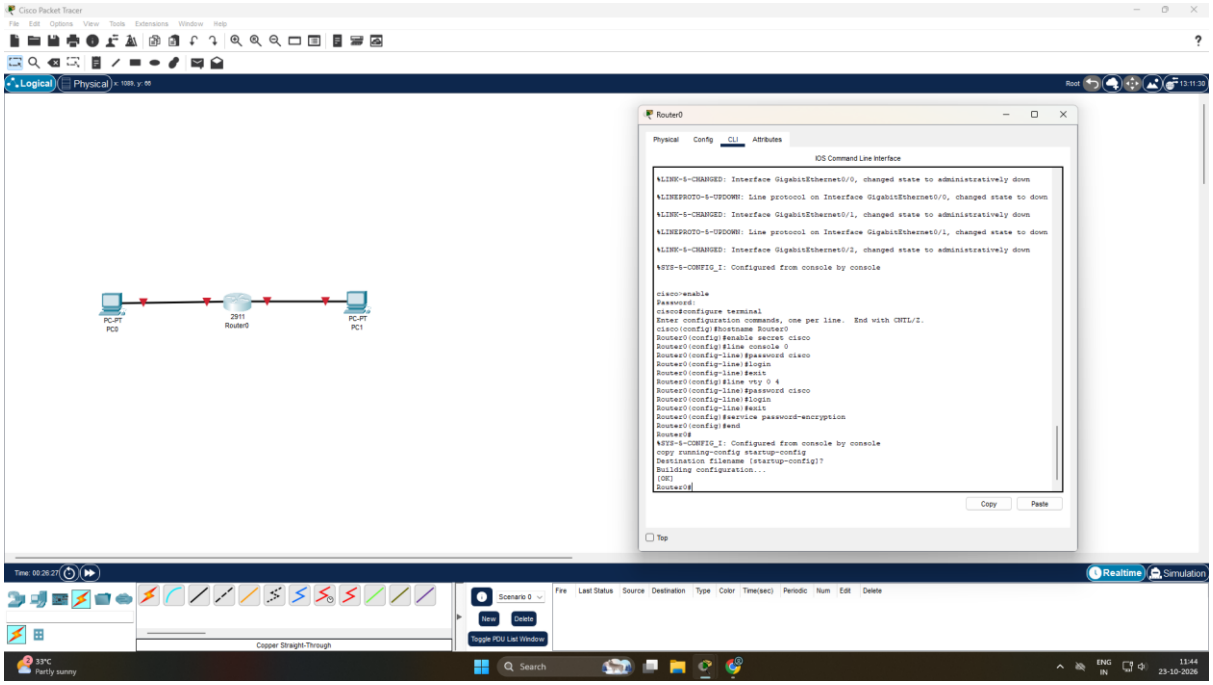
The CLI displayed the message:

Building configuration...

[OK]

This confirms that the configuration was saved to the startup configuration.

Screenshot:



8. Conclusion

Basic configuration of the Cisco 2911 router was successfully completed using Cisco IOS CLI commands. Passwords were configured for privileged EXEC, console, and virtual terminal access. Password encryption was enabled, and the configuration was saved successfully.